

Hyperuniformity in Chemotactic Active Matter

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1 Models

1.1 Particle-based model

$$\frac{d}{dt} \mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D} \boldsymbol{\xi}(t) , \quad (1a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + k \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) , \quad (1b)$$

for $i = 1, 2, \dots, N$. Here, $\mathbf{r}_i$ is the position of the i -th particle, c is the concentration field of the chemoattractant, α is the chemotactic sensitivity. If $\alpha > 0$, the particles are attracted to higher concentrations of the chemoattractant (chemoattraction/positive chemotaxis), while if $\alpha < 0$, the particles are repelled by higher concentrations of the chemoattractant (chemorepulsion/negative chemotaxis). D is the diffusion coefficient of the particles, $\boldsymbol{\xi}$ is a Gaussian white noise with zero mean and unit variance, D_c is the diffusion coefficient of the chemoattractant, μ is the decay ($\mu < 0$) or growth ($\mu > 0$) rate of the chemoattractant, and k is the production ($k > 0$) or consumption rate ($k < 0$) of the chemoattractant by the particles.

For simplicity, we assume that particles are point-like and move in a two-dimensional $L \times L$ square domain with periodic boundary conditions. The initial positions of the particles can be randomly distributed, and the initial concentration field can be set to a small random perturbation around a uniform background concentration c_0 .

To reduce the parameter space, we can non-dimensionalize the equations by introducing $x_0 = L$, $t_0 = 1/k$ and c_0 as characteristic scales of length, time and concentration. The resulting dimensionless parameters are $\tilde{\alpha} = \frac{\alpha c_0}{L^2 k}$, $\tilde{D} = \frac{D}{L^2 k}$, $\tilde{D}_c = \frac{D_c}{L^2 k}$, and $\tilde{\mu} = \frac{\mu}{k}$. We can omit the tildes and work with the dimensionless equations:

$$\frac{d}{dt} \mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D} \boldsymbol{\xi}(t) , \quad (2a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (2b)$$

Note that the above equations mathematically implies the assumption of $k > 0$. If $k < 0$, the non-dimensionalization would be different, and the Eq. (2b) would have a negative sign in front of the summation term:

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) - \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (3)$$

In the following, we will focus on the case of $k < 0$ (chemorepulsion/negative chemotaxis), which is more relevant to the emergence of hyperuniformity.

1.2 Continuum model

It is convenient to describe collective behavior using a particle density field $\rho(\mathbf{r}, t) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i(t))$. The continuum model can be derived from the particle-based model using a mean-field approximation, which assumes that the particles are uniformly distributed and that the fluctuations around the mean density are negligible. The resulting equations are:

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho, \quad (4a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho. \quad (4b)$$

Note that the $\rho(\mathbf{r}, t)$ here is particle number density, not probability density.